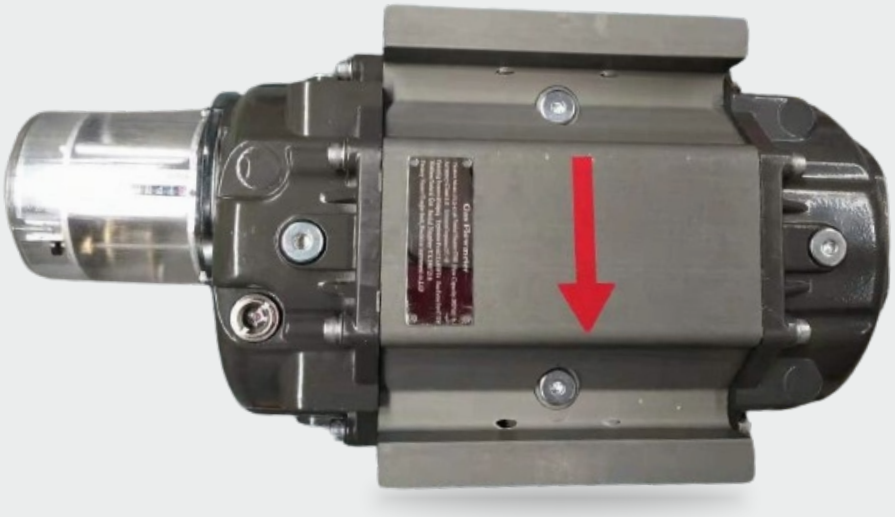
IOTAFLOW
SYSTEMS PVT. LTD

GasPro RPD Gas meter

Mechanical head, without temperature and pressure compensation, can show volume flow under actual working conditions (m³/h)



1. Structure and working principle

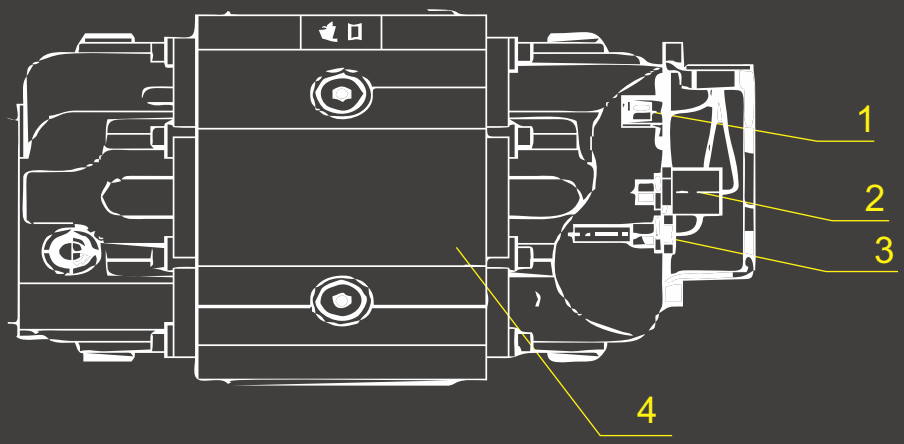


FIG. 1 schematic diagram of flow meter structure

1, flow sensor 2, pressure sensor 3, temperature sensor 4, gas waist wheel flow meter

2. Working principle of intelligent GasPro RPD Gas meter (see figure 2)

The mechanical reader GasPro RPD Gas meter is mainly composed of shell, conjugate rotor and mechanical reader. A pair of conjugate rotors installed in the metering chamber keep the rotor in correct relative position through precision machined adjusting gears under the action of inlet and outlet pressure difference of the circulating gas (P into > P out). Keep the best working clearance between rotor, between rotor and shell, between rotor and wall board, realize the continuous non-contact seal. For each rotation of the rotor, the gas with four times the effective volume of the measuring chamber will be output. The rotor revolution will be transmitted to the mechanical reader through the magnetic seal coupling device and deceleration mechanism, so as to display the cumulative volume of the output gas.

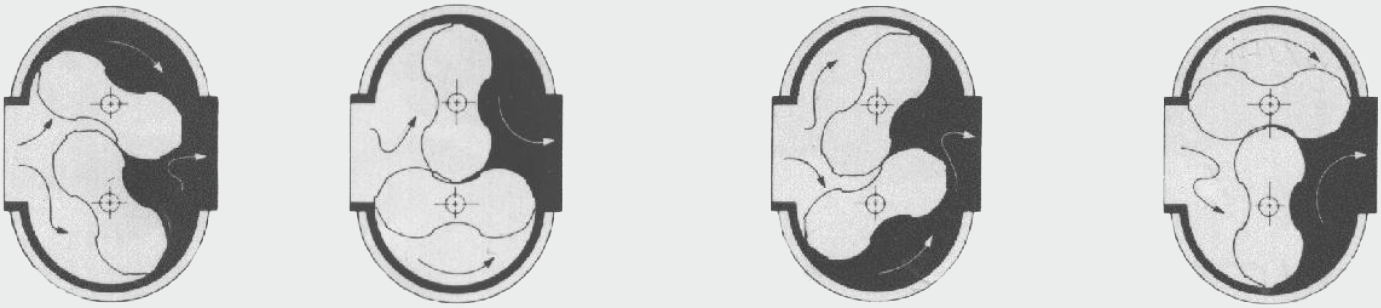


FIG. 2 working principle of GasPro RPD Gas meter

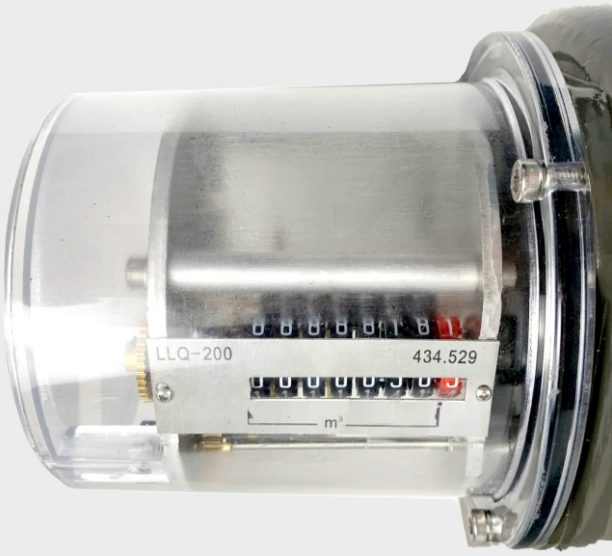
3. Main technical parameters and functions

3.1 flowmeter specifications, basic parameters and performance indicators (see table 1)

Nominal diameter DN(mm)	Upper Limit of flow m 3/h	Lower Limit of flow m 3/h	Pressure Loss at QMAX pa	Nominal Pressure Mpa	Precision %	Rangeability	Material
25	20	1	100	1.0	1.0 1.5	20:1	Aluminum alloy
40	30	1.5	100			30 :1	
40	40	2	1 40			40:1	
50	6 0	3	100			60 :1	
50	85	4	1 40			70 :1	
80	100	6	1 10			70 :1	
80	1 4 0	8	150			120:1	
80	200	8	180			100:1	
100	2 5 0	10	180			110:1	
100	300	10	130			110 :1	
100	450	20	200			11 0 :1	
150	650	30	35 0			80 :1	Aluminum
150	1000	40	450			80:1	Aluminum
200	1600	60	50 0			80 :1	Aluminum

The nominal pressure below 1.0mpa is aluminum alloy material, and above 1.6mpa is cast steel material or stainless steel material or cast steel aluminum alloy combined profile material.

3.2 Standard condition： P=101.325 kPa T= 293.15K



3.3 Conditions of use：

a) Environment temperature：－25 ℃～+80℃

b) Medium temperature：－20 ℃～+60℃

c) Relative humidity：5% ～95%

d) The atmospheric pressure：86KPa～106Kpa

3.4 Electrical performance index：

3.4.1 Working power supply：

a) The power supply ：One section of 3VDC lithium battery, the battery voltage in 2.0v ~ 3.0v can work normally.
Replace the battery when the voltage is below 2.5v.

b) external power supply：±24VDC±15%, ripple ≤± 5%, suitable for 4 ~ 20mA output, pulse output, RS-485, etc.

3.4.2 Power consumption: external power supply, ≤1W 。
Internal power supply, average power consumption ≤1mw, can work for more than five years continuously.

3.4.3 Working condition of the pulse output

a) Working condition pulse signal: the working condition pulse signal detected by the flowsensor is amplified and output directly through optical coupler isolation, with high level 20V and low voltage level ≤ 1V.The pulse is mainly used to calibrate the instrument

b) Frequency signals proportional to the standard volume flow are amplified and output by optical coupler isolation, with high level 20V and low level ≤1V.

c) Calibration pulse signal, output amplitude 0 ~ 3V, output pulse width 500ms.The pulse is mainly used for city gas meters IC card control unit or other equipment. The signal 1m³
outputs 1 pulse.

3.4.4 (4 ~ 20) mA standard analog current output function
Is proportional to the standard volume flow, 4mA corresponds to 0 m³ /h, and 20mAcorresponds to the maximum standard volume flow (this value can be set).

3.4.5 RS485 communication (photoelectric isolation RS485 communication module is adopted)
3.4.5 RS485 communication (using photoelectric isolation RS485 communication module)
This integrator can connect with PC and PLC through the RS485 standard interface, and carry on serial communication. Display medium pressure, temperature, instantaneous flow, accumulative standard flow, battery voltage.

4. Selection and installation

4.1 In order to improve the accuracy of measurement, it is necessary for the intelligent gas GasPro RPD Gas meter to convert the working flow into the standard flow.
Standard state (20 ℃) a standard atmospheric pressure, temperature of 2 0 degrees Celsius under volume flow, can press the type conversion：

$$Q_{standard} = \frac{273.15+20}{101325} \times \frac{P}{T} \times Q$$

In the formula：$Q_{standard}$ -- Volume flow under standard conditions (Nm^3/h)
$Q_{working}$ -- The volume flow rate under actual working conditions at the time (m^3/h)

P --The actual absolute pressure of the gas in the pipeline at that time.
If the pressure fluctuates in the metering process, it is the actual average absolute pressure of the gas in the pipeline (Pa).

$P = P_{working} + B$
$P_{working}$ -- The average gauge pressure of a gas read by a manometer at that time (Pa)

B -- Local atmospheric pressure（Pa）
T -- If the actual absolute temperature (K) of the gas in the pipeline at that time fluctuates during the metering process, it is the actual average absolute pressure of the gas in the pipeline
T=t+273.15K
t -- The average temperature of a gas in centigrade as measured by the thermometer（℃）

4.2 External dimensions and installation of flowmeter (see figure 3 and table 2)

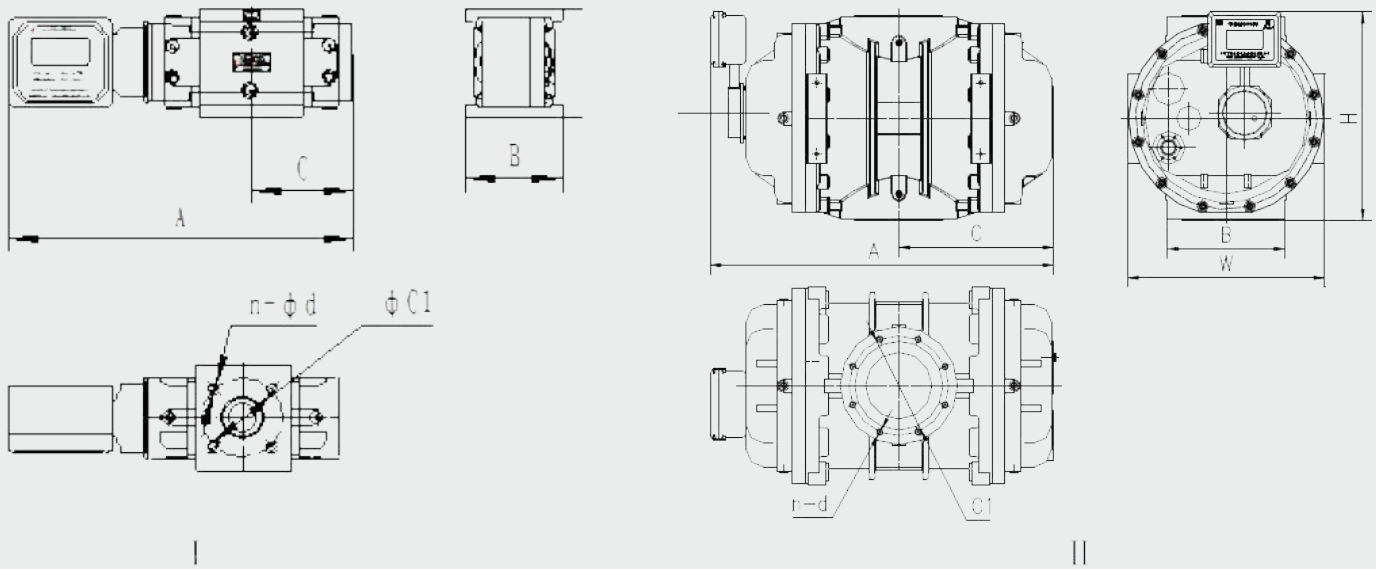


figure 3 Installation size of flow meter

4.2.1 Installation dimensions of flow meter (see table 2)

Nominal diameter DN	Direction of inlet and outlet	A (mm)	B (mm)	C (mm)	W (mm)	H (mm)	Structure and form	Weight (kg)	Flange	
									C1	n-d
25	Top down out	409	135	107	135	144	Ⅰ、Ⅱ	9	ϕ75	4-M10
40	Top down out	433	135	119	135	144	Ⅰ、Ⅱ	11	ϕ90	4-M12
40	Top down out	463	135	134	135	144	Ⅰ、Ⅱ	13	ϕ100	4-M12
50	Top down out	470	180	137.5	180	172	Ⅰ、Ⅱ	15	ϕ125	4-M12
50	Top down out	507	180	156	180	172	Ⅰ、Ⅱ	16.5	ϕ125	4-M12
80	Top down out	539	180	172	180	172	Ⅰ、Ⅱ	18	ϕ160	8-M16
80	Top down out	575	180	190	180	172	Ⅰ、Ⅱ	20	ϕ160	8-M16
80	Top down out	569	210	187	210	245	Ⅰ、Ⅱ	24	ϕ160	8-M16
100	Top down out	647	210	226	210	245	Ⅰ、Ⅱ	30	ϕ180	8-M16
100	Top down out	647	210	226	210	245	Ⅰ、Ⅱ	30	ϕ180	8-M16
100	Top down out	743	210	274	210	245	Ⅰ、Ⅱ	36.5	ϕ180	8-M16
150	Top down out	745	ϕ285	275	370	380	Ⅰ、Ⅱ	92.5	ϕ240	8-M16
150	Top down out	779	ϕ285	292	460	490	Ⅰ、Ⅱ	111	ϕ240	8-M16
200	Top down out	875	ϕ340	340	460	510	Ⅰ、Ⅱ	125	ϕ295	8-M20

4.3.2 Installation of flow meter

a) Before installation and use, users should read this instruction carefully to ensure the correct installation and reasonable use of the flowmeter, so as not to affect the normal operation and measuring accuracy.

b) Before selecting the flowmeter, the proper model and specification should be correctly selected according to the pressure, flow rate, temperature of the measured gas and the inlet and outlet orientation required by the process flow (see"flow conversion" for calculation method).

c) Before installing the flowmeter, the inlet and outlet package should be removedand granular impurities must be prevented from falling into the measuring room. If the measuring room is coated with anti-rust oil, it can be rinsed with gasoline or kerosene, and impurities in the pipe should be strictly removed. A filter or a filterscreen should be installed upstream of the flowmeter to prevent rust, welding slag and other impurities from entering the measuring room

d) When the flowmeter is installed, whether the inlet and outlet are in the vertical or horizontal position, the rotor axis should be kept as horizontal as possible.

e) When the gas pressure fluctuates within a large range, a voltage regulator shall be installed on the upstream of the flowmeter to ensure measurement accuracy.

f) In order to prevent rust slag, welding slag and other impurities in the newly installed pipeline from entering into the flowmeter, the user should first install the transition pipe at the installation position of the flowmeter, ventilate it for a period of time, and then replace it with the flowmeter after ensuring there are no large particle impurities. Installation flow timing, should ensure that the flow meter center and pipeline center alignment, no dislocation, and the flow meter is not affected by external forces (including axial and radial).Such as pipeline welding, should be installed in excess pipe, not directly with the flow meter welding.

4.4 Rationalization proposal It is suggested to install the flowmeter vertically. The flow direction of the gas is upper and lower flow. Under special circumstances, it can adopt lower and upper flow or left and right flow. Installation flow timing should be set before and after the valve and bypass pipeline, to ensure that maintenance, do not have to stop gas. Figure 4, figure 5 and figure 6 are respectively the typical system drawings of installation pipelines of three flowmeters: A, Band C (the inlet and outlet directions are upper and lower, lower and upper, left and right respectively), for the reference of users.

4.5 The size of distribution pipes and fittings in each part of the installation pipeline of theflowmeter must be appropriate, and the flowmeter body shall not be subjected to abnormal external forces.

4.6 After the installation of the flowmeter, inject the lubricating oil (GB486~65 high-speedoil H5) from the filling port to the middle line of the oil window (not more).
Check regularly or irregularly during use to ensure sufficient lubricating oil and cleanliness. If found in the use of oil blackening or oil level Higher than the middle line of oil window, it indicates that the lubricating oil is deteriorating or has impurities, at this time should be replaced with a new lubricating oil.If the oil level is3mm below the center line of the oil window.
It indicates that the lubricating oil loss should be supplemented to the middle line of oil window. When filling lubricating oil, it is necessary to close the valve before and after the flowmeter, empty the air in the flowmeter, and then fill lubricating oil.

4.6.1 When the flowmeter is put into operation, all valves should be opened slowly to prevent the strong impact of airflow from damaging the flowmeter.

4.6.2 After the flowmeter is put into operation for a period of time, if the filter pressure drop is found to increase, the filter should be cleaned or replaced. If the pressure drop of the flowmeter increases and the starting flow rate increases, clean gasoline or kerosene can be used to flush the measuring room.

4.6.3 When the flowmeter is out of service for a long time, the lubricating oil should be emptied and the measuring room cleaned with gasoline. After that, the inlet and outlet should be sealed and placed in a dry place.

Gas volume correctors

ELCOR and ELCORplus

Highlights

- High accuracy and stability of measurement
- Embedded modem 2G/3G/4G/NB
- Robust design
- Modular technology
- MID and ATEX approved
- Long battery life
- Designed for the future upgrades
- Tamper proof

Introduction

The latest lotaflow technology ELCOR plus and ELCOR represent powerful solution applicable either like electronic volume corrector for billing purposes or like universal datalogger for station control and monitoring.

Both models are designed as compact devices integrating measuring metrological part together with communication unit and with telemetry features as well.

ELCORplus

The most powerfull model is ELCORplus. Its robust design together with tremendous modularity covers all known demands of gas segment over EVC (ELCORplus) and datalogger installation (DATCOMplus).

Battery powered ELCORplus converts gas volume under operating conditions to gas volume at base conditions, according to state equation. Information about primary volume can be received from all types of gasmeters through LF pulses (reed contact or wiegand sensor), HF pulses and absolute encoders (NAMUR, SCR+). The meter can be connected through mechanical index with lotaflow relative encoder. EVC calculates the conversion factor C based on actual value of gas pressure, gas temperature and based on compressibility factor according to all known methods.

The basic concept enables to use device either as a single or dual channel EVC with possibility to add non- metrological channel. Modular architecture enables to create various configurations up to 6 pressure or temperature transducers, 12 digital inputs, 4 digital outputs, 6 analogue inputs and 4 analogue outputs, based on customer requirements.

Any configuration can be individually customised by parameter file locally or remotely.

All data (instantaneous, archived data, diagnostics) can be visible on backlit graphical LCD display with using 6-buttons keypad. This keypad and LCD combination enables simple parameters setup as well.

All measured and calculated data can be optionally stored at 11 different archieves with sufficient memory capacity covering couple of years without overwriting of old records.

In addition to metrological measuring part ELCORplus can handle two independent communication modems. internal and external. There are selectable modem modules for 2G, 3G, 4G and 4G NB. Modular design is suitable for easy replacement with future generation of moderns without interruption of functioning. All versions even with internal modem are approved for installation in hazardous areas Ex Zone 0, 1, 2.

Lifetime of modem battery with regular calling once a day is guaranteed at least 6 years in defined mode of working. For more frequent communication (or permanent) the external power supply module (AC or solar) can be applied for powering EVC and modem separately. Except modems there are paralelly usable two selectable serial interfaces RS232 and RS485 and local infrared interface.

The data can be transferred independently to data collection system (billing) or monitoring and control system (Scada), Various communication protocols are available. The protocol can be selected by enduser for each communication channel independently. The data transfer is equipped with access protection, encryption tools and system of certified firmware remote download. ELCORplus is designed in robust polycarbonate housing suitable for indoor or outdoor installation. This cabinet is fully resistant against external tampering and disturbances with massive locking lever.

ELCOR

ELCOR is simplified version of bigger ELCORplus.
The versatility is limited comparing the bigger model. It is designed as single channel EVC with only one RS485 (or RS232) serial port and with optional embedded modem both powered by batteries. ELCOR is suitable for standard projects where reliable and smart EVC is required.

Datalogger DATCOM and DATCOMplus

For telemetry projects where dataloggers are required DATCOM and DATCOMplus are available. They use identical HW and FW features as EVC series enhanced with functionality typical for dataloggers installations. Advantages of open architecture together with flexible internal structure of IOTAFLOW dataloggers series bring unlimited solution for any project in gas segment.

Technical Specification

	ELCOR	ELCORplus
Housing	The housing was designed for use in outdoor with focus on robustness and resistance against tampering. It can be sealed and secured by a suitable lock. The housing is equipped with a closing mechanism keeping the lid closed tightly even without securing the lid by screws	
Material	Glass fiber polycarbonate	
Dimensions	210 x 190 x 95 mm	263 x 201 x 11 mm
Weight (depends on configuration	1.5 kg	2.2 kg
Protection class (EN 60529)	IP 66, outdoor environment	
Working temperature	-25°C to +70°C (optionally from-40°C)	
Control panel	The new ELCOR series brings even better capacitive keypad very well known from picoELCOR series.	
	6 button capacitive keypad	
Display	Local reading of all measured instantaneous and archived data.	
Standard variant	Graphic backlit LCD 128 x 04	
Optional	Alphanumeric segment backlit CD (working from -40 °C, always ON)	
Digital transducers	The new generation of digital transducers offers traditional high stability and accuracy of IOTAFLOW pressure and temperature transducers.	
	Up to 2 digital transducers	Up to 6 digital transducers
Pressure transducer type	EDT 96	
Standard MID approved ranges-absoluts (accuracy 0.25% of measured value)	3.8-6.2:2-10:4 20:7 35:14 70:25 130 bor	
Extended MID approved ranges-abeolote (accuracy 0.25% of measured value)	08-10:0.8-20; 4-70 bar	
Non MID ranges- absolute (accuracy 0.2% full scale)	0.8-35; 0.8-70:0.8-130 bar	
Non MID ranges gauge (accuracy 0.4% full scale - below 1 bar) (accuracy 1.2% full scale-above 1 bar)	0-020-10-1.60-40-80-100-25:0-40.0-70 bar	
Temperature transducer type	EDT 101	
Measuring range standard	-25°C to +70°C	
Measuring range extended	-40°C to +70°C	

Power supply	The device is designed as battery power supplied (ELCOR) with an option of external power supply (ELCORplus) either from solar panel or from mains 120V/230V AC (using external modules placed in non-hazardous area)	
EVC battery packs- battery life (in defined conditions)	B-03 (17Ah) khium battery - up to 11 years	B-03 (17Ah) lithium battery up to 11 years B-03D (30Ah) lithium battery up to 18 years B-03A alkaline battery-up to 6 years
Modem battery packs- battery life (in defined conditions)	HB-03 (GSM/GPRS modem, lithium battery. 12Ah) - up to 6 years HB-03D (GSM/GPRS modem, lithium battery, 20Ah - up to 11 years HB-01 (LTE modem, lithium battery, 12Ah) up to 6 years	
External power supply	not available	PS-E external moxiule (EVC) PS-M external modulu (MODEM)

Communication with the device	Both models supports various communication interfaces allowing local or remote communication.	
Serial communication interface	1x RS232 or R5465	2x RS232 or Rs486
Optical communication interface	optical interface (EN 52056-21)	
Wireless communication (modem)	GSM/GPRS LTE Cat1 LTE M1 LTE NB (preliminary)	
Inputs/Outputs	The device provides digital inputs and outputs fully configurable by end user.	
Digital inputs	3x (LF or binary)	up to 12x (LF, 2x HF, binary)
Digital outputs	2x (LF or binary)	4x (LF or binary)
Analog inputs (4-20mA)	not available	up to 6x (optional)
Analog outputs (4-20mA)	2x (using external module CLO)	up to 4x integrated (optional) up to 4x (using external module CLO)

Encoder interface	Various encoders are supported
Encoder types supported	NAMUR SCR+ IOTAFLOW

Extension internal modules		ELCOR and ELCORplus series was developed as modular device enabling full customization.	
Communication	**Moderns**	M2G: 2G modem M4G: 2G/3G/4G - LTE Cat1 modem M4GM1: LTE M1 modem LTE-NB (preliminary	
	Serial interface	integrated	S1-COM: RS232/RS485 serial interface
Gas meter encoder interface		S0-SCR: SCR+ encoder interface SO-NAM-NAMUR.encoder interface.	
Analog 4-20mA Input/output	**Inputs**	not available	S1-2AI: 2x input
	Outputs	not available (externally via CLO module)	S1-2AO: 2x output
External power supply	**EVC**	not available	S3-PWR1
	Modem		S4-PWR2
Multipurpose extension board		not available	S0-EXT1: 2x HF NAMUR input 6x LF/Binary digital input 2x Analog input (4-20 mA) 2x Digital transducer EDT

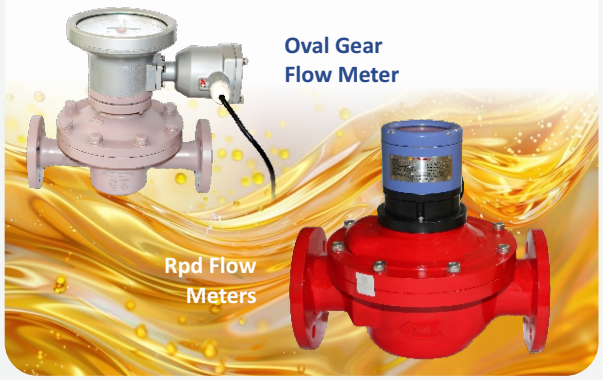
Beyond the Best: Explore More

WATER FLOW METER

Ideal for measuring liquid flow with high accuracy and minimal pressure drop. The perfect choice for a wide range of industrial applications

POSITIVE DISPLACEMENT OIL METERS

Highly accurate for measuring viscous liquids. Perfect for industries like oil, chemical, and food processing where precise volume measurements are essential.

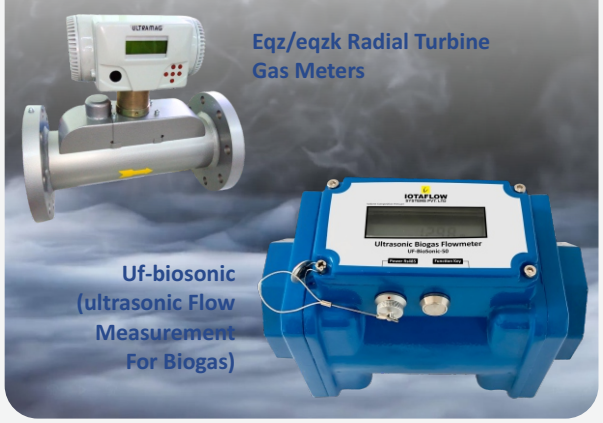


AIR FLOW METERS

Designed to monitor and control air flow in HVAC systems, compressed air systems, and more. Ensures efficiency and accurate flow measurement in air handling processes.

GAS METERS

These meters ensure precise measurement of natural gas, biogas, and other gases, enhancing safety and operational efficiency.



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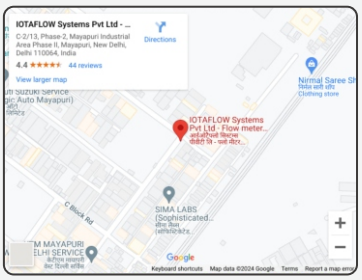

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